Nishal Stanislaus Silva, *PhD*

Machine Learning Research Engineer | Real-Time & Edge Inference, C++17 | Embedded ML

Canadian Permanent Resident, Eligible to work in Canada without sponsorship

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Re: Engineer, AI Software/Machine Learning (C++) – Qualcomm

Dear Hiring Manager,

I am applying for the Engineer, AI Software/Machine Learning (C++) role at Qualcomm. I am a machine learning research engineer with a PhD from the University of Trento (2025) and 10+ years across industry and academia, and my work is real-time inference under tight latency and compute budgets on embedded ARM hardware. I built real-time inference services on a Raspberry Pi 4 that run in under 30ms at over 90% F1, and I released Nebula, an open-source C++17 feature-extraction library for sequential time series data.

On the C++ side, Nebula is written in C++17 and benchmarked per feature on ARM, and at MAS Holdings I wrote custom C++ for a printer integration that coordinated three closed-loop subsystems in real time. On the ML side, I have trained and deployed models in TensorFlow and PyTorch, and I have used ONNX. I work day to day with Linux, Git and Docker, and I have written CI/CD and code review practices for teams.

I have not written Hexagon DSP kernels or vector-instruction code, and I have not worked with the Hexagon SDK. What I bring is the habit of profiling first, measuring on the target device, and making a design fit its latency budget, and I would learn the Hexagon toolchain quickly. Qualcomm AI Engine Direct is the layer where model performance on real devices is decided, and I would like to work on the operations and tooling that make it fast. In my first months I would aim to contribute optimized C++17 operations and to debug customer use cases.

I am a Canadian permanent resident based in Toronto, eligible to work without sponsorship, and available immediately. I would welcome the opportunity to discuss how my experience in real-time ML systems and embedded C++ can support the AI Software team at Qualcomm. My portfolio, publications and open datasets are available at nishal.xyz.

Sincerely,

Nishal Stanislaus Silva, PhD